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== Vitis HLS Report for 'matVecMul'
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* Date: Fri Jan 1 00:00:19 2021

* Version: 2021.1 (Build 3247384 on Thu Jun 10 19:36:07 MDT 2021)
* Project: matVecMul_stream_prj
* Solution: solution_extrachocolate (Vivado IP Flow Target)
* Product family: zynq
* Target device: xc7z020-clg400-1

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== Performance Estimates
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+ Timing:
    * Summary:

Clock	Target	Estimated	Uncertainty
ap_clk	10.00 ns	81.734 ns	2.70 ns

+ Latency:
    * Summary:

Latency (cycles) min	Latency (cycles) max	Latency (absolute) min	Latency (absolute) max	Interval min	Interval max	Pipeline Type
1031	21881	84.268 us	1.788 ms	1032	21882	no

    + Detail:
        * Instance:

Instance	Module	Latency (cycles) min	Latency (cycles) max	Latency (absolute) min	Latency (absolute) max	Interval min	Interval max	Pipeline Type
grp_matVecMul_Pipeline_sample_loop_fu_481	matVecMul_Pipeline_sample_loop	66	66	5.394 us	5.394 us	66	66	no
grp_matVecMul_Pipeline_matmul_inner_loop_fu_628	matVecMul_Pipeline_matmul_inner_loop	10	11	0.817 us	0.899 us	10	11	no

        * Loop:

Loop Name	Latency (cycles) min	Latency (cycles) max	Iteration Latency	Initiation Interval achieved	Initiation Interval target	Trip Count	Pipelined
- vector_loop	1030	21880	1030 ~ 1094	-	-	1 ~ 20	no
+ matmul_outer_loop	960	1024	15 ~ 16	-	-	64	no

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== Utilization Estimates
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* Summary:

Name	BRAM_18K	DSP	FF	LUT	URAM
DSP	-	-	-	-	-
Expression	-	-	0	407	-
FIFO	-	-	-	-	-
Instance	32	48	8498	13516	-
Memory	-	-	-	-	-

Multiplexer	-	-	-	107	-
Register	-	-	3514	-	-
Total	32	48	12012	14030	0
Available	280	220	106400	53200	0
Utilization (%)	11	21	11	26	0

+ Detail:
    * Instance:

Instance	Module	BRAM_18K	DSP	FF	LUT	URAM
control_s_axi_U	control_s_axi	32	0	1430	1352	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_628	matVecMul_Pipeline_matmul_inner_loop	0	48	7034	12002	0
grp_matVecMul_Pipeline_sample_loop_fu_481	matVecMul_Pipeline_sample_loop	0	0	34	97	0
mux_164_70_1_1_U206	mux_164_70_1_1	0	0	0	65	0
Total		32	48	8498	13516	0

    * DSP:
    N/A

    * Memory:
    N/A

    * FIFO:
    N/A

    * Expression:

Variable Name	Operation	DSP	FF	LUT	Bitwidth P0	Bitwidth P1
add_ln44_fu_791_p2	+	0	0	39	32	1
i_2_fu_775_p2	+	0	0	38	31	1
p_Val2_1_fu_1257_p2	+	0	0	39	32	32
sub22_fu_752_p2	+	0	0	39	32	2
sub_fu_746_p2	+	0	0	39	32	2
and_ln789_fu_1343_p2	and	0	0	2	1	1
and_ln790_fu_1357_p2	and	0	0	2	1	1
and_ln795_fu_1390_p2	and	0	0	2	1	1
carry_1_fu_1277_p2	and	0	0	2	1	1
out_stream_TLAST_int_regslice	and	0	0	2	1	1
overflow_fu_1384_p2	and	0	0	2	1	1
underflow_fu_1400_p2	and	0	0	2	1	1
Range1_all_ones_fu_1309_p2	icmp	0	0	14	22	2
Range1_all_zeros_fu_1315_p2	icmp	0	0	14	22	1
Range2_all_ones_fu_1293_p2	icmp	0	0	14	21	2
cmp21_fu_781_p2	icmp	0	0	18	32	32
icmp_ln32_fu_770_p2	icmp	0	0	18	32	32
icmp_ln44_fu_786_p2	icmp	0	0	18	32	32
icmp_ln69_fu_1428_p2	icmp	0	0	18	32	32
or_ln384_fu_1414_p2	or	0	0	2	1	1
or_ln794_fu_1374_p2	or	0	0	2	1	1
deleted_ones_fu_1349_p3	select	0	0	2	1	1
deleted_zeros_fu_1321_p3	select	0	0	2	1	1
out_stream_TDATA_int_regslice	select	0	0	32	1	32
select_ln384_fu_1406_p3	select	0	0	33	1	31
xor_ln416_fu_1271_p2	xor	0	0	2	1	2
xor_ln789_fu_1337_p2	xor	0	0	2	1	2
xor_ln790_fu_1363_p2	xor	0	0	2	2	1
xor_ln794_1_fu_1379_p2	xor	0	0	2	1	2
xor_ln794_fu_1369_p2	xor	0	0	2	1	2
xor_ln795_fu_1394_p2	xor	0	0	2	2	1

Total		0	0	407	373	255

* Multiplexer:

Name	LUT	Input Size	Bits	Total Bits
ap_NS_fsm	53	10	1	10
ap_phi_mux_neg_src_1_phi_fu_474_p4	9	2	1	2
i_fu_342	9	2	31	62
in_stream_TREADY_int_regslice	9	2	1	2
k_reg_458	9	2	32	64
neg_src_1_reg_470	9	2	1	2
out_stream_TDATA_blk_n	9	2	1	2
Total	107	22	68	144

* Register:

Name	FF	LUT	Bits	Const Bits
acc_V_0_loc_fu_406	70	0	70	0
add_ln44_reg_1584	32	0	32	0
ap_CS_fsm	9	0	9	0
cmp21_reg_1576	1	0	1	0
col_size_read_reg_1446	32	0	32	0
deleted_ones_reg_1812	1	0	1	0
deleted_zeros_reg_1807	1	0	1	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_628_ap_start_reg	1	0	1	0
grp_matVecMul_Pipeline_sample_loop_fu_481_ap_start_reg	1	0	1	0
i_2_reg_1571	31	0	31	0
i_fu_342	31	0	31	0
k_reg_458	32	0	32	0
neg_src_1_reg_470	1	0	1	0
no_of_vectors_read_reg_1457	32	0	32	0
p_Result_2_reg_1801	1	0	1	0
p_Result_s_reg_1791	1	0	1	0
p_Val2_1_reg_1796	32	0	32	0
row_size_read_reg_1452	32	0	32	0
select_ln340_10_loc_fu_386	70	0	70	0
select_ln340_11_loc_fu_390	70	0	70	0
select_ln340_12_loc_fu_394	70	0	70	0
select_ln340_13_loc_fu_398	70	0	70	0
select_ln340_14_loc_fu_402	70	0	70	0
select_ln340_1_loc_fu_350	70	0	70	0
select_ln340_2_loc_fu_354	70	0	70	0
select_ln340_3_loc_fu_358	70	0	70	0
select_ln340_4_loc_fu_362	70	0	70	0
select_ln340_5_loc_fu_366	70	0	70	0
select_ln340_6_loc_fu_370	70	0	70	0
select_ln340_7_loc_fu_374	70	0	70	0
select_ln340_8_loc_fu_378	70	0	70	0
select_ln340_9_loc_fu_382	70	0	70	0
select_ln340_loc_fu_346	70	0	70	0
shl_ln_reg_1589	6	0	10	4
sub22_reg_1563	32	0	32	0
sub_reg_1558	32	0	32	0
targetBlock_reg_1786	4	0	4	0
x_localbuff_V_0_0	32	0	32	0
x_localbuff_V_0_1	32	0	32	0
x_localbuff_V_0_2	32	0	32	0
x_localbuff_V_0_3	32	0	32	0
x_localbuff_V_10_0	32	0	32	0
x_localbuff_V_10_1	32	0	32	0
x_localbuff_V_10_2	32	0	32	0

x_localbuff_V_10_3	32	0	32	0
x_localbuff_V_11_0	32	0	32	0
x_localbuff_V_11_1	32	0	32	0
x_localbuff_V_11_2	32	0	32	0
x_localbuff_V_11_3	32	0	32	0
x_localbuff_V_12_0	32	0	32	0
x_localbuff_V_12_1	32	0	32	0
x_localbuff_V_12_2	32	0	32	0
x_localbuff_V_12_3	32	0	32	0
x_localbuff_V_13_0	32	0	32	0
x_localbuff_V_13_1	32	0	32	0
x_localbuff_V_13_2	32	0	32	0
x_localbuff_V_13_3	32	0	32	0
x_localbuff_V_14_0	32	0	32	0
x_localbuff_V_14_1	32	0	32	0
x_localbuff_V_14_2	32	0	32	0
x_localbuff_V_14_3	32	0	32	0
x_localbuff_V_15_0	32	0	32	0
x_localbuff_V_15_1	32	0	32	0
x_localbuff_V_15_2	32	0	32	0
x_localbuff_V_15_3	32	0	32	0
x_localbuff_V_1_0	32	0	32	0
x_localbuff_V_1_1	32	0	32	0
x_localbuff_V_1_2	32	0	32	0
x_localbuff_V_1_3	32	0	32	0
x_localbuff_V_2_0	32	0	32	0
x_localbuff_V_2_1	32	0	32	0
x_localbuff_V_2_2	32	0	32	0
x_localbuff_V_2_3	32	0	32	0
x_localbuff_V_3_0	32	0	32	0
x_localbuff_V_3_1	32	0	32	0
x_localbuff_V_3_2	32	0	32	0
x_localbuff_V_3_3	32	0	32	0
x_localbuff_V_4_0	32	0	32	0
x_localbuff_V_4_1	32	0	32	0
x_localbuff_V_4_2	32	0	32	0
x_localbuff_V_4_3	32	0	32	0
x_localbuff_V_5_0	32	0	32	0
x_localbuff_V_5_1	32	0	32	0
x_localbuff_V_5_2	32	0	32	0
x_localbuff_V_5_3	32	0	32	0
x_localbuff_V_6_0	32	0	32	0
x_localbuff_V_6_1	32	0	32	0
x_localbuff_V_6_2	32	0	32	0
x_localbuff_V_6_3	32	0	32	0
x_localbuff_V_7_0	32	0	32	0
x_localbuff_V_7_1	32	0	32	0
x_localbuff_V_7_2	32	0	32	0
x_localbuff_V_7_3	32	0	32	0
x_localbuff_V_8_0	32	0	32	0
x_localbuff_V_8_1	32	0	32	0
x_localbuff_V_8_2	32	0	32	0
x_localbuff_V_8_3	32	0	32	0
x_localbuff_V_9_0	32	0	32	0
x_localbuff_V_9_1	32	0	32	0
x_localbuff_V_9_2	32	0	32	0
x_localbuff_V_9_3	32	0	32	0
xor_ln790_reg_1817	1	0	1	0
Total	3514	0	3518	4

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== Interface
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* Summary:

RTL Ports	Dir	Bits	Protocol	Source Object	C Type
s_axi_control_AWVALID	in	1	s_axi	control	array
s_axi_control_AWREADY	out	1	s_axi	control	array
s_axi_control_AWADDR	in	15	s_axi	control	array
s_axi_control_WVALID	in	1	s_axi	control	array
s_axi_control_WREADY	out	1	s_axi	control	array
s_axi_control_WDATA	in	32	s_axi	control	array
s_axi_control_WSTRB	in	4	s_axi	control	array
s_axi_control_ARVALID	in	1	s_axi	control	array
s_axi_control_ARREADY	out	1	s_axi	control	array
s_axi_control_ARADDR	in	15	s_axi	control	array
s_axi_control_RVALID	out	1	s_axi	control	array
s_axi_control_RREADY	in	1	s_axi	control	array
s_axi_control_RDATA	out	32	s_axi	control	array
s_axi_control_RRESP	out	2	s_axi	control	array
s_axi_control_BVALID	out	1	s_axi	control	array
s_axi_control_BREADY	in	1	s_axi	control	array
s_axi_control_BRESP	out	2	s_axi	control	array
ap_clk	in	1	ap_ctrl_hs	matVecMul	return value
ap_rst_n	in	1	ap_ctrl_hs	matVecMul	return value
interrupt	out	1	ap_ctrl_hs	matVecMul	return value
in_stream_TDATA	in	32	axis	in_stream_V_data_V	pointer
in_stream_TVALID	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TREADY	out	1	axis	in_stream_V_dest_V	pointer
in_stream_TDEST	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TKEEP	in	4	axis	in_stream_V_keep_V	pointer
in_stream_TSTRB	in	4	axis	in_stream_V_strb_V	pointer
in_stream_TUSER	in	1	axis	in_stream_V_user_V	pointer
in_stream_TLAST	in	1	axis	in_stream_V_last_V	pointer
in_stream_TID	in	1	axis	in_stream_V_id_V	pointer
out_stream_TDATA	out	32	axis	out_stream_V_data_V	pointer
out_stream_TVALID	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TREADY	in	1	axis	out_stream_V_dest_V	pointer
out_stream_TDEST	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TKEEP	out	4	axis	out_stream_V_keep_V	pointer
out_stream_TSTRB	out	4	axis	out_stream_V_strb_V	pointer
out_stream_TUSER	out	1	axis	out_stream_V_user_V	pointer
out_stream_TLAST	out	1	axis	out_stream_V_last_V	pointer
out_stream_TID	out	1	axis	out_stream_V_id_V	pointer